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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

Digital Note Organizer with File Storage

Submitted by

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25BCS145

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DECLARATION

I hereby declare that the project entitled "**Digital Note Organizer with File Storage**" submitted in partial fulfillment of the requirements for the course **Java programming language is a record of my original work** carried out under the guidance of MR.**Aravind sir**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

Santhanalakshmi N

CERTIFICATE

This is to certify that the project entitled **"Digital Note Organizer with File Storage"** is a **bona fide work** carried out by **Santhanalakshmi N,** CS-**I'B'-2025**, **Department Computer Science**, **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **JAVA PROGRAMMING LANGUAGE**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: Coimbatore

Date:10/04/2026

Submitted for the Viva-Voce Examination held on 10/04/2026

Guide:	**Head of the Department**
(Technical Trainer, KGCAS)	(Technical Head, KGCAS)
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ABSTRACT

The **Digital Note Organizer with File Storage** is a software application designed to provide an efficient, reliable, and user-friendly solution for managing notes and digital files in a centralized system. In today's digital era, individuals such as students, teachers, and professionals generate a large amount of information in the form of notes, documents, images, and PDFs. However, traditional methods of note-taking and file management often lead to issues such as misplacement, lack of organization, difficulty in retrieval, and absence of proper security.

The system is designed using Object-Oriented Programming (OOP) principles, which enhance modularity, reusability, and scalability of the application. Additional concepts such as file handling and database connectivity (JDBC) are used to manage and store data effectively. The graphical user interface (GUI) is developed to ensure ease of use, making the system accessible even to users with minimal technical knowledge.

In conclusion, the Digital Note Organizer with File Storage provides a practical and efficient solution for modern note management needs.

1. INTRODUCTION

In today's fast-growing digital era, the need for efficient information management has become increasingly important. With the rapid growth of technology, individuals such as students, teachers, researchers, and professionals deal with a large volume of data in the form of notes, documents, images, and other digital files on a daily basis. Managing this information using traditional methods like paper notebooks or unorganized digital storage often leads to several challenges, including data loss, poor organization, and difficulty in retrieving required information at the right time.

The **Digital Note Organizer with File Storage** is developed to address these issues by providing a structured and user-friendly platform for managing digital content. This system enables users to create, edit, delete, and store notes efficiently, while also allowing them to upload and manage various types of files such as PDFs, images, and documents. By combining note-taking and file storage into a single application, the system ensures better accessibility and improved organization of information.

The application is developed using Java, a powerful and platform-independent programming language. It utilizes Object-Oriented Programming (OOP) concepts such as classes, objects, inheritance, and polymorphism to ensure modularity, scalability, and code reusability. Additionally, features like file handling and database connectivity (JDBC) are incorporated to manage and store user data securely and efficiently. The system also includes a graphical user interface (GUI), which makes it easy for users to interact with the application without requiring advanced technical knowledge.

1.1 Objective

The main objective of the Digital Note Organizer with File Storage is to develop an efficient and user-friendly system for managing notes and digital files in an organized and secure manner. The project focuses on improving productivity and reducing the difficulties faced in traditional note management systems. Specific Objectives:

- To develop a digital platform for creating, storing, and managing notes
- To provide a centralized system for storing various types of files such as documents, images, and PDFs
- To reduce the dependency on physical notebooks and paper-based systems
- To enable users to easily create, edit, and delete notes
- To implement a search feature for quick retrieval of stored notes and files

1.2 Overview

The **Digital Note Organizer with File Storage** is a software application designed to help users efficiently manage their notes and digital files in a centralized and organized manner. In today's fast-paced digital environment, individuals handle a large amount of information daily, including text notes, documents, images, and PDFs. Managing these resources using traditional methods or scattered digital storage can be inefficient and time-consuming.

The application is developed using Java and is based on Object-Oriented Programming (OOP) principles, ensuring modularity and scalability. It also uses file handling and database connectivity (JDBC) to store and manage data efficiently. A user-friendly graphical interface (GUI) is designed to allow smooth interaction between the user and the system.

Overall, the Digital Note Organizer with File Storage provides a reliable, secure, and easy-to-use platform for managing digital content. It is suitable for students, teachers, and professionals and can be further enhanced with advanced features such as cloud storage, mobile access, and improved security mechanisms.

1.3 Features

The **Digital Note Organizer with File Storage** provides several useful features to ensure efficient and organized management of notes and files.

Main Features:

- **User-Friendly Interface**
  Simple and easy-to-use design for all types of users
- **Create, Edit, and Delete Notes**
  Users can easily manage their notes anytime
- **File Upload and Storage**
  Supports storing files like PDFs, documents, and images
- **Centralized Storage System**
  All notes and files are stored in one place
- **Search Functionality**
  Quickly find notes using keywords
- **Categorization of Notes**
  Notes can be organized into different categories or folders
- **Secure Data Management**
  Ensures safety of stored information
- **Quick Access and Retrieval**
  Users can access their data instantly
- **Time-Saving System**

2. PROGRAM ANALYSIS

Program analysis is an essential phase in software development where the problem is studied in detail and the requirements of the system are identified. In this project, The proposed system, **"Digital Note Organizer with File Storage",** is designed to overcome these limitations by providing a computerized solution. It allows users to create, edit,
Overall, the program analysis helps in designing a system that improves efficiency, reduces manual effort, enhances accuracy, and provides a reliable solution for managing digital notes and files. It lays a strong foundation for the successful development and implementation of the project.

2.1 Problem Definition

In today's digital world, managing notes and important files is an essential task for students, teachers, and professionals. However, many people still rely on traditional methods such as physical notebooks or unorganized digital storage, which leads to several difficulties.

Some common problems include:

- Loss of important notes and files
- Poor organization of data

The **Digital Note Organizer with File Storage** is designed to overcome these problems by providing a computerized solution

2.2 Project Overview

The **Digital Note Organizer with File Storage** is a Java-based application developed to simplify the process of managing notes and digital files in an organized manner.

The system provides a user-friendly interface where users can create, edit, and store notes, This project is implemented using Core Java programming concepts and can be developed as either a console-based program or a Java Swing GUI application. It uses features like Object-Oriented Programming (OOP), file handling, and database connectivity (JDBC) to ensure efficient performance and data management

2.3 Scopes and Limitations

Scope

The scope of the **Digital Note Organizer with File Storage** includes the following:

- Creating and managing digital notes
- Editing and deleting existing notes
- Uploading and storing files such as PDFs, images, and documents
- Organizing notes and files in a structured manner
- Searching and retrieving notes quickly
- Providing centralized storage for all data.

3. SYSTEM DESIGN

System design is the stage where the structure and components of the system are planned before the actual implementation. In this phase, the system is divided into different modules, and the workflow of the application is clearly defined

The **Digital Note Organizer with File Storage** is designed as a user-friendly application that allows users to manage their notes and files in a centralized system. The application provides options to create, edit, delete, and store notes digitally, along with uploading and managing files such as documents, images, and PDFs. The system ensures that all data is organized in a structured format, making it easy for users to access and retrieve information whenever required.

3.1 Modules

The main modules of the Digital Note Organizer with File Storage are explained below:

1. Menu Display Module

This module is responsible for displaying the list of available notes along with their titles or categories. It helps the user to view and select the required notes easily.

2. Note Management Module

This module is responsible for handling all operations related to creating, updating, and deleting notes in the system. It is one of the core modules of the Digital Note Organizer, as it manages the main functionality of note handling.

3. File Storage Module

This module is responsible for handling the uploading, storing, and managing of files within the system. It plays a key role in allowing users to keep all their important documents along with their notes in a single organized platform.

4. User Module

This module is responsible for handling user interactions with the system. It acts as the interface between the user and the application, allowing users to access and use the features of the Digital Note Organizer efficiently.

5.Data Storage

This module is responsible for storing and managing all the data in the system, including notes and uploaded files. It ensures that the information entered by the user is saved properly and can be retrieved whenever required**Module**

4.METHODOLOGY

Methodology describes the process followed to develop the project. It explains the steps taken from understanding the problem to designing, implementing, testing, and documenting the system.

For the Digital Note Organizer with File Storage, a structured development approach was followed to ensure that the system meets user requirements and functions efficiently.

The development process includes the following stages:

- Problem Analysis
- System Design
- Program Development
- Testing and Validation
- Result and Documentation

4.1. Problem Analysis

In this phase, the problem is studied and analyzed in detail. The requirements of the system are identified based on the difficulties faced in managing notes and digital files using traditional methods.

The main problems identified in existing note management systems are:

- Notes and files can be easily lost or misplaced.
- Lack of proper organization leads to confusion.

Based on these problems, the system requirements were defined to develop a digital application that simplifies note management and file storage by providing an organized, efficient, and user-friendly solution.

4.2. System Design

In this stage, the overall structure of the system is designed before coding begins.

The system is divided into modules such as:

- User Module
- Note Display Module
- Note Management Module
- File Storage Module
- Search Module

The program flow and structure are designed so that users can easily interact with the system to create, manage, and retrieve notes and files efficiently.

4.3. Program Development

After designing the system, the program is implemented using the Java programming language.

The following steps were followed during development:

- Creation of classes and methods
- Implementation of note management logic (add, edit, delete)
- Writing code for file upload and storage
- Handling user input and interactions
- Implementing search functionality for notes and files
- Displaying stored notes and file details

Core Java concepts such as classes, objects, loops, conditional statements, file handling, and methods were used during development.

4.4. Testing and Validation

Testing ensures that the program works correctly and produces the expected results.

Different test cases were used to check the functionality of the system.

Example Test Cases:

Test Case Expected Result

Create note Note is saved and displayed correctly

Edit note Updated note is shown correctly

Delete note Note is removed from the list

Upload file File is stored successfully

Search note Relevant results are displayed

Invalid input Error handled properly

This phase ensures that the program runs without errors and manages notes and files accurately an

This phase ensures that the program runs without errors and manages notes and files accurately and efficiently.

4.5. Result and Documentation

After successful testing, the results of the system are documented. The system allows users to create, store, manage, and retrieve notes along with uploading and accessing files efficiently.

The results are demonstrated through program execution and screenshots showing the creation of notes, file uploads, search functionality, and the display of stored data.

Proper documentation is prepared to explain the design, implementation, and working of the system, including details of modules, features, and technologies used.

4.6. Algorithm

Algorithm for the **Digital Note Organizer with File Storage**:

Step 1: Start the program

Step 2: Initialize storage for notes and files **Step 3:** Display the main menu options:

- Create Note
- View Notes
- Edit Note
- Delete Note

- Upload File
- Search Note/File
- Exit

Step 4: Read the user's choice

Step 5: If the user selects **Create Note**

- Enter note title and content
- Save the note

Step 6: If the user selects **View Notes**

- Display all saved notes

Step 7: If the user selects **Edit Note**

- Select a note
- Modify the content
- Save the updated note

Step 8: If the user selects **Delete Note**

- Select a note
- Delete the note

Step 9: If the user selects **Upload File**

- Choose a file from the system
- Store the file

Step 10: If the user selects **Search Note/File**

- Enter a keyword
- Display matching results

Step 11: If the user selects **Exit**

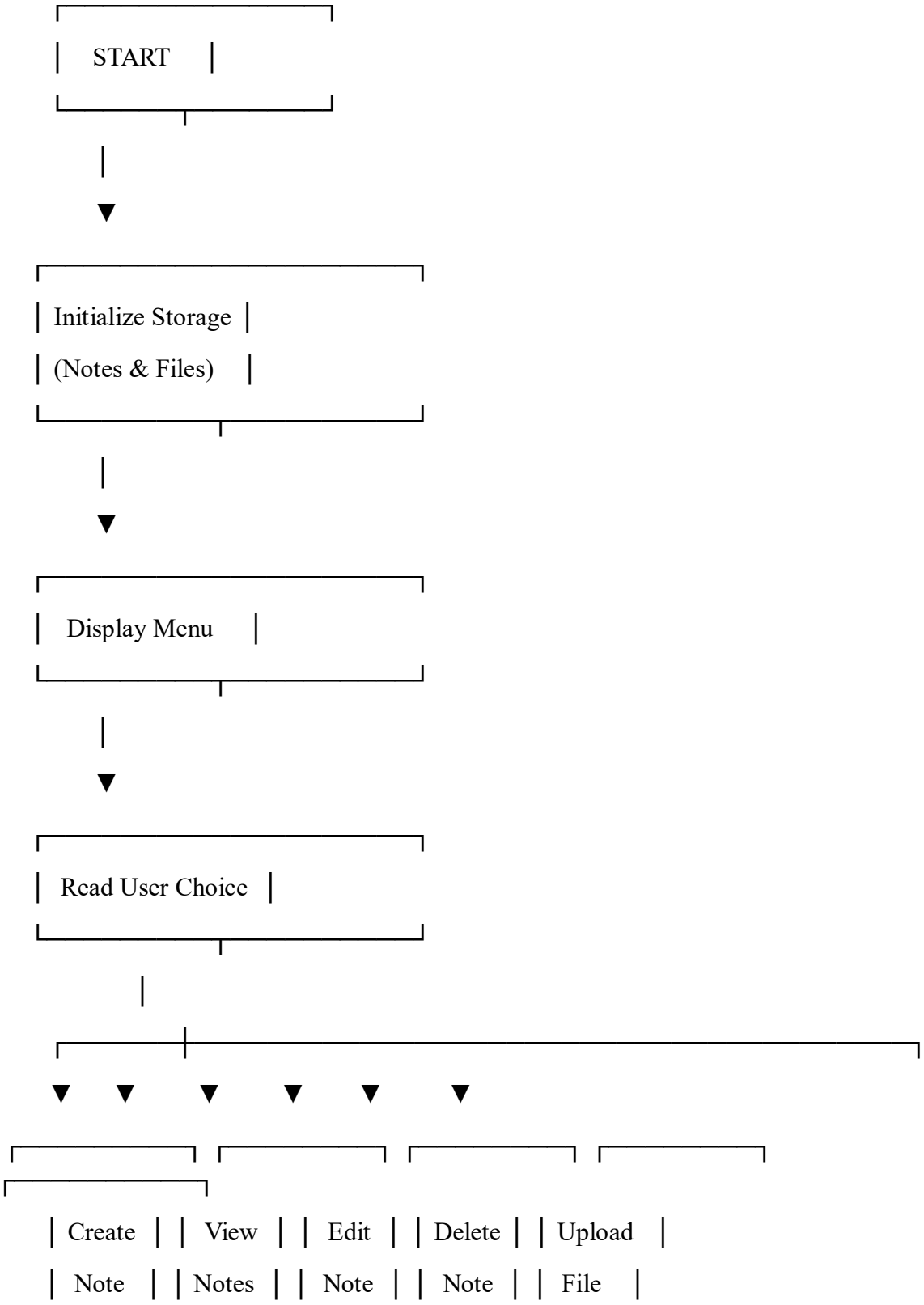
- Terminate the program

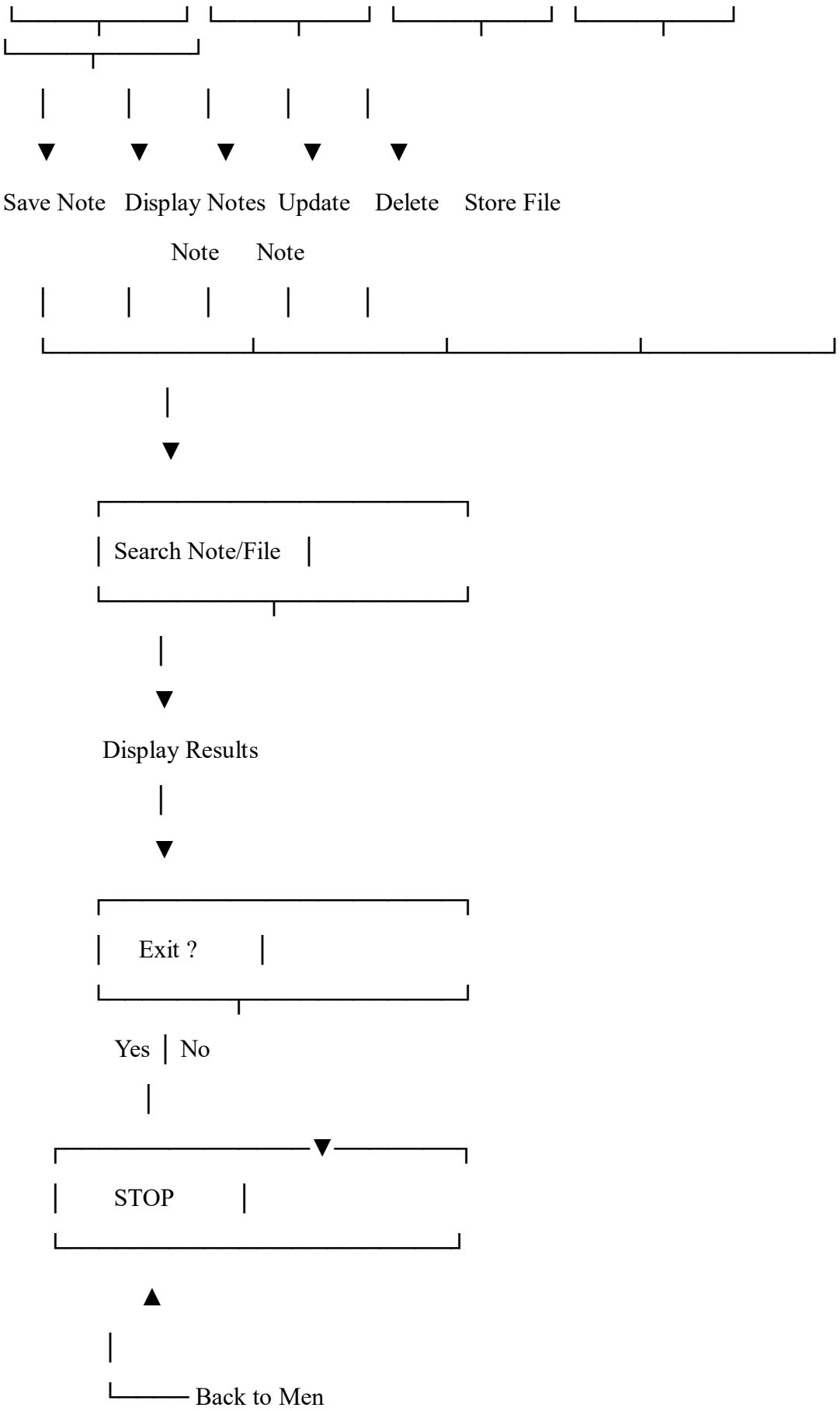
Step 12: If the choice is invalid .Display an error message

Step 13: Repeat steps 3 to 12 until the user exits

Step 14: Stop

4.7. Flowchart




Save Note
Display Notes
Update
Delete
Store File
Note
Note
Search Note/File
Display Results
Exit ?
Yes
No
STOP
Back to Men

5. SOURCE CODE

```
import javax.swing.*;
import javax.swing.table.DefaultTableModel;
import java.awt.*; import java.awt.event.*;
import java.io.*;

public class DigitalNoteOrganizer extends JFrame {

    JLabel titleLabel, contentLabel;
    JTextField titleField;
    JTextArea contentArea;
    JButton addBtn, saveBtn, loadBtn, deleteBtn;
    JTable table;
    DefaultTableModel model;

    public DigitalNoteOrganizer() {        setTitle("Digital
Note Organizer");        setSize(700, 500);
setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
setLayout(new BorderLayout());



        JPanel topPanel = new JPanel(new GridLayout(2, 2));

        titleLabel = new JLabel("Note Title:");
titleField = new JTextField();        contentLabel =
new JLabel("Note Content:");        contentArea =
new JTextArea();
```

```
        topPanel.add(titleLabel);
topPanel.add(titleField);
topPanel.add(contentLabel);         topPanel.add(new
JScrollPane(contentArea));

        add(topPanel, BorderLayout.NORTH);



        model = new DefaultTableModel();
model.addColumn("Title");

        table = new JTable(model);         add(new
JScrollPane(table), BorderLayout.CENTER);



        JPanel btnPanel = new JPanel();

        addBtn = new JButton("Add Note");
saveBtn = new JButton("Save to File");
loadBtn = new JButton("Load File");
deleteBtn = new JButton("Delete Note");

        btnPanel.add(addBtn);
btnPanel.add(saveBtn);         btnPanel.add(loadBtn);
btnPanel.add(deleteBtn);         add(btnPanel,
BorderLayout.SOUTH);

        addBtn.addActionListener(e -> {
String title = titleField.getText();            if
```

```
(!title.isEmpty()) {
model.addRow(new Object[]{title});
titleField.setText("");
contentArea.setText("");
        } else {
            JOptionPane.showMessageDialog(this, "Enter title!");
        }
    });

    saveBtn.addActionListener(e -> {
        JFileChooser fileChooser = new JFileChooser();
int option = fileChooser.showSaveDialog(this);

        if (option == JFileChooser.APPROVE_OPTION) {
            try {
                File file = fileChooser.getSelectedFile();
                BufferedWriter writer = new BufferedWriter(new FileWriter(file));

                for (int i = 0; i < model.getRowCount(); i++) {
writer.write(model.getValueAt(i, 0).toString());                    writer.newLine();
                }
writer.close();
                JOptionPane.showMessageDialog(this, "Saved Successfully!");
            } catch (Exception ex) {
ex.printStackTrace();
            }
        }
    });
    loadBtn.addActionListener(e -> {
```

```
        JFileChooser fileChooser = new JFileChooser();
int option = fileChooser.showOpenDialog(this);

        if (option == JFileChooser.APPROVE_OPTION) {
            try {
                File file = fileChooser.getSelectedFile();
                BufferedReader reader = new BufferedReader(new FileReader(file));

                model.setRowCount(0);
                String line;

                while ((line = reader.readLine()) != null) {
model.addRow(new Object[]{line});
                }
                 reader.close();
                JOptionPane.showMessageDialog(this, "Loaded Successfully!");
            } catch (Exception ex) {
ex.printStackTrace();
            }
        }
    });

    deleteBtn.addActionListener(e -> {
int row = table.getSelectedRow();          if
(row >= 0) {
model.removeRow(row);
        } else {
            JOptionPane.showMessageDialog(this, "Select a note to delete!");
        }
```

```
        });

        setVisible(true);
    }

    public static void main(String[] args) {
new DigitalNoteOrganizer();
    }
}
```

6. INPUT AND OUTPUT SCREENS

Figure 1: Input Screen

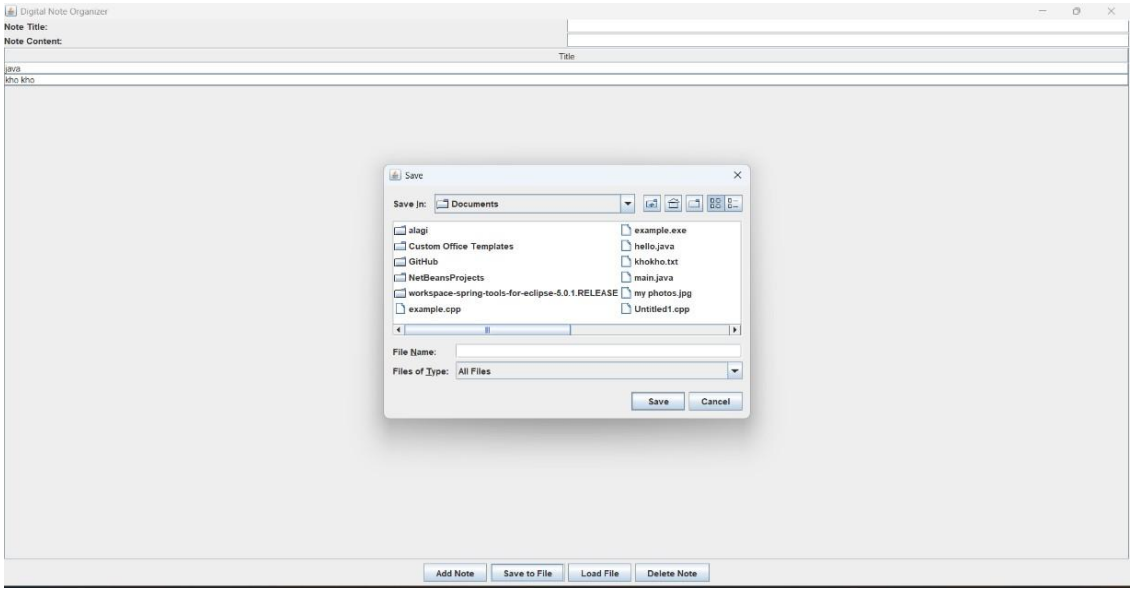
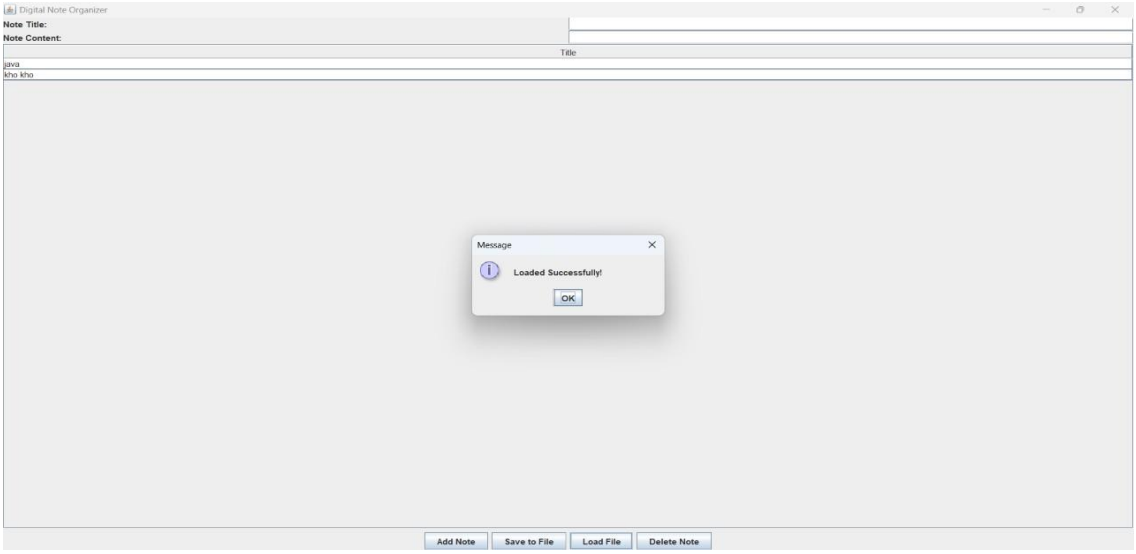


Figure 2: Output Screen



7. DISCUSSION & CONCLUSION

7.1 Discussion

The **Digital Note Organizer with File Storage** was developed to simplify the process of managing notes and storing files in a digital format. The system allows users to create, edit, delete, and view notes, as well as upload and manage files efficiently.

During the development process, Core Java programming concepts such as loops, conditional statements, methods, file handling, and object-oriented principles were applied. The system successfully organizes notes and stores files in a structured and accessible manner.

The system was tested with different inputs to ensure that all functionalities such as note creation, editing, deletion, file upload, and search operations work correctly. The results show that the system reduces manual effort, improves organization, and provides quick access to stored information.

7.2 Conclusion

The **Digital Note Organizer with File Storage** successfully demonstrates the practical application of Core Java programming concepts in developing a useful real-world application.

The system simplifies note management by allowing users to create, edit, delete, and organize notes efficiently. It also enables file storage, making it easier to manage both notes and documents in a single platform. This reduces manual effort and improves accessibility of information.

This project helps students understand how Java programming can be used to build practical software systems for everyday use. The project can be further enhanced by adding features such as cloud storage, advanced security, graphical user interfaces, and multi-user access.

8. REFERENCES

The following resources were used during the development of this project:

Books

• Herbert Schildt – *Java: The Complete Reference*

Websites

• Official Java Documentation – https://docs.oracle.com
• TutorialsPoint – https://www.tutorialspoint.com/java
• W3Schools Java Tutorial – https://www.w3schools.com/java